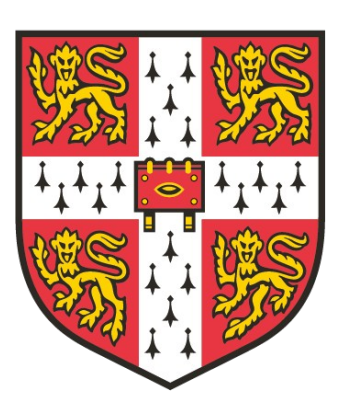




UNIVERSITY OF
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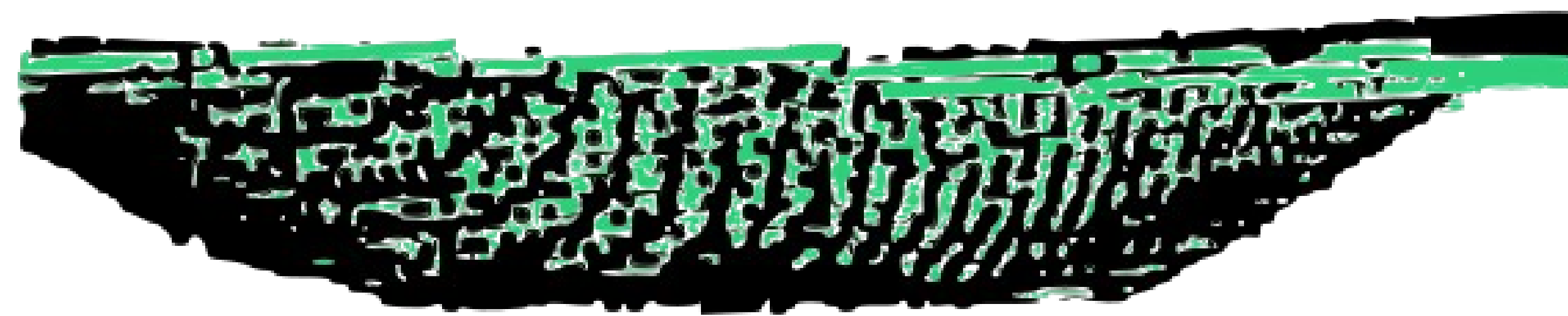


TERRA-ROBOTICS LAB

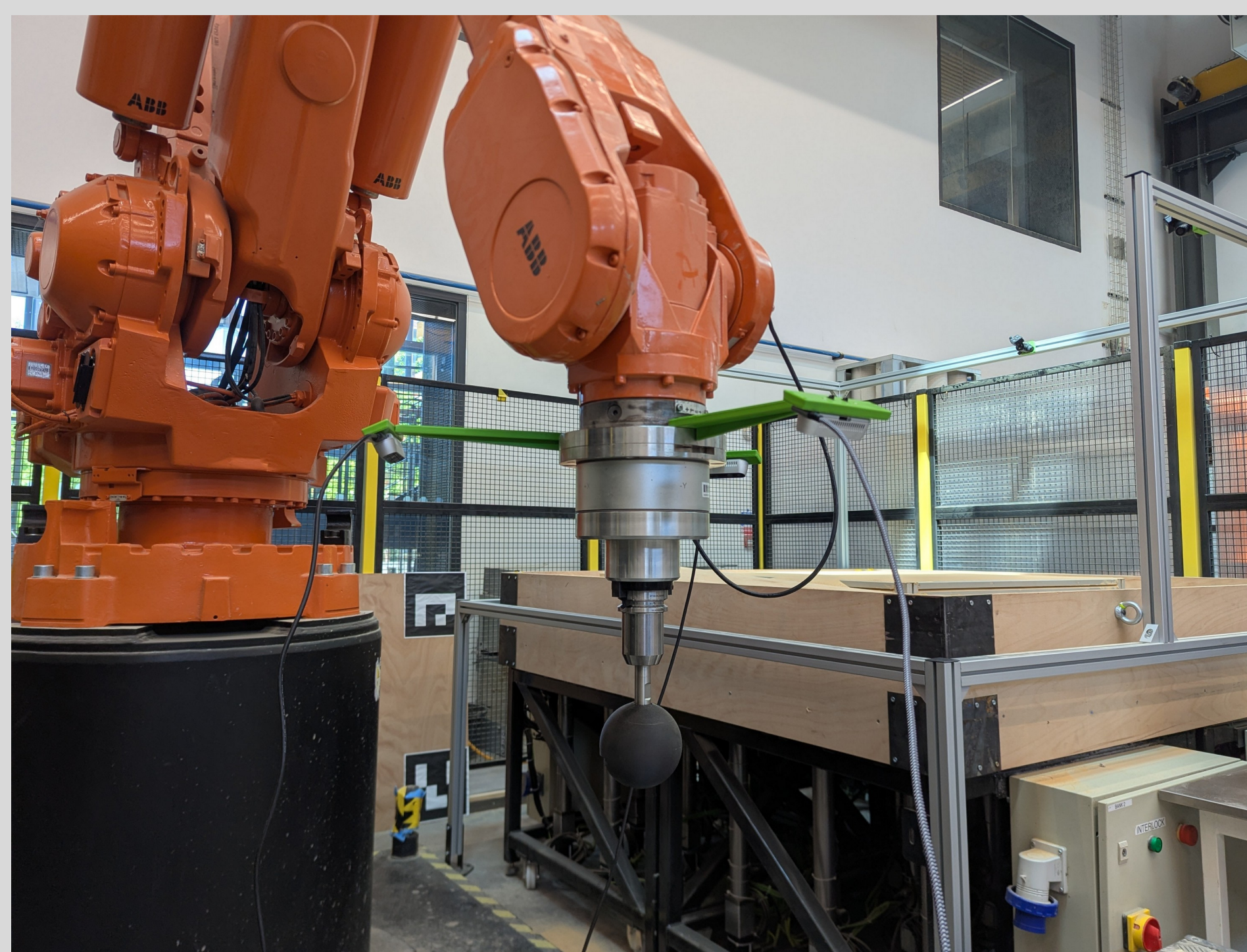


J.Ingham – come find me!

Eye-in-hand computer vision for autonomous geotechnical assessment – MaGIC 2026



Side view: Accurate and Inaccurate point cloud registration for a soil indent – (Black: Industry standard LiDAR, G/R: RGBD camera)



Terra Robotics Laboratory robot and multi-RGBD setup

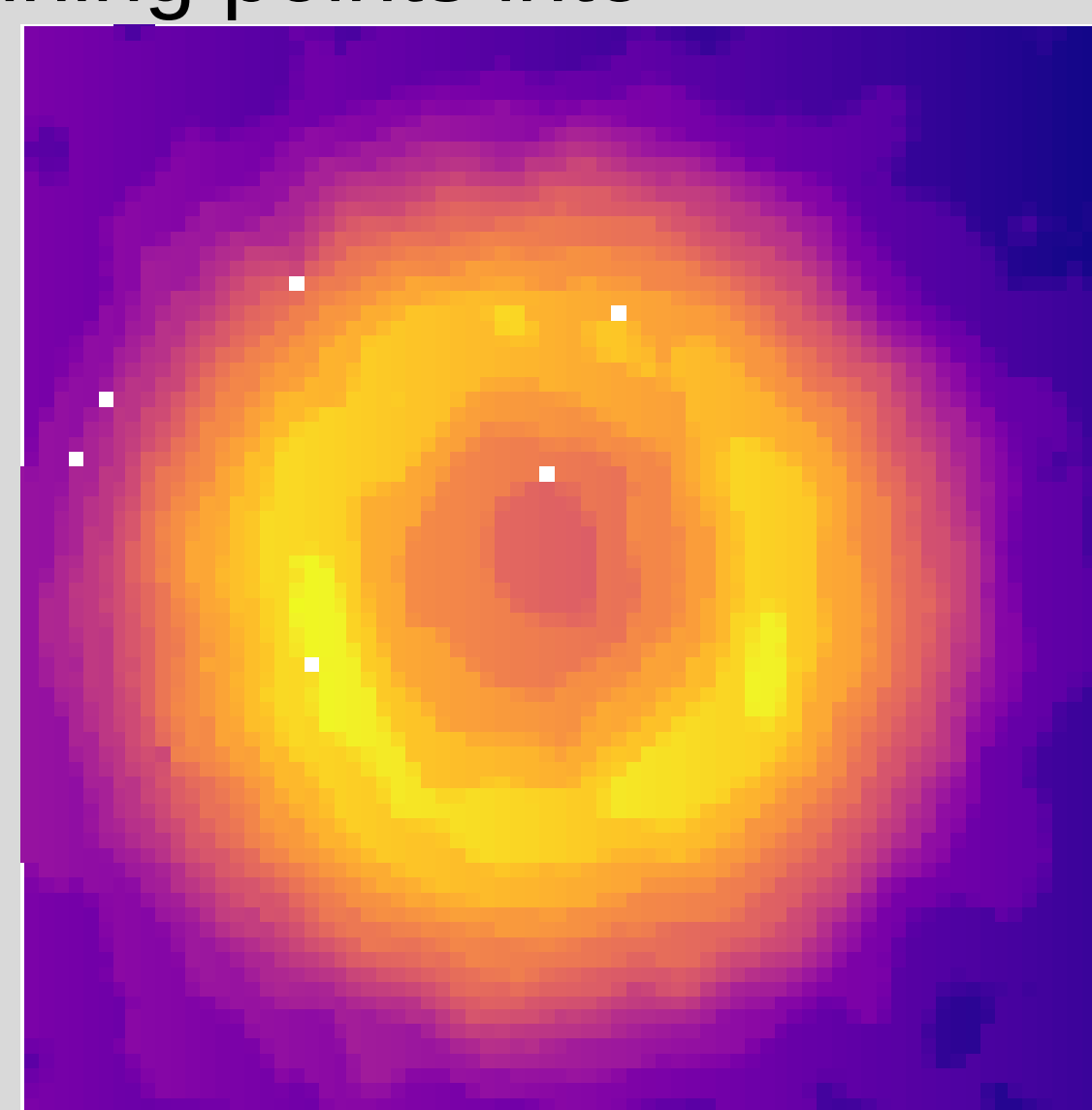
Motivation

- Autonomous earth moving requires knowledge of earth state (visually)
- Lack of earth moving datasets
- Can cheap RGBD cameras replace expensive industry-standard LiDAR?
- Can point clouds be compressed and still retain information?

Heightmaps

- Heightmaps are produced by vertically binning points into a grid-based format

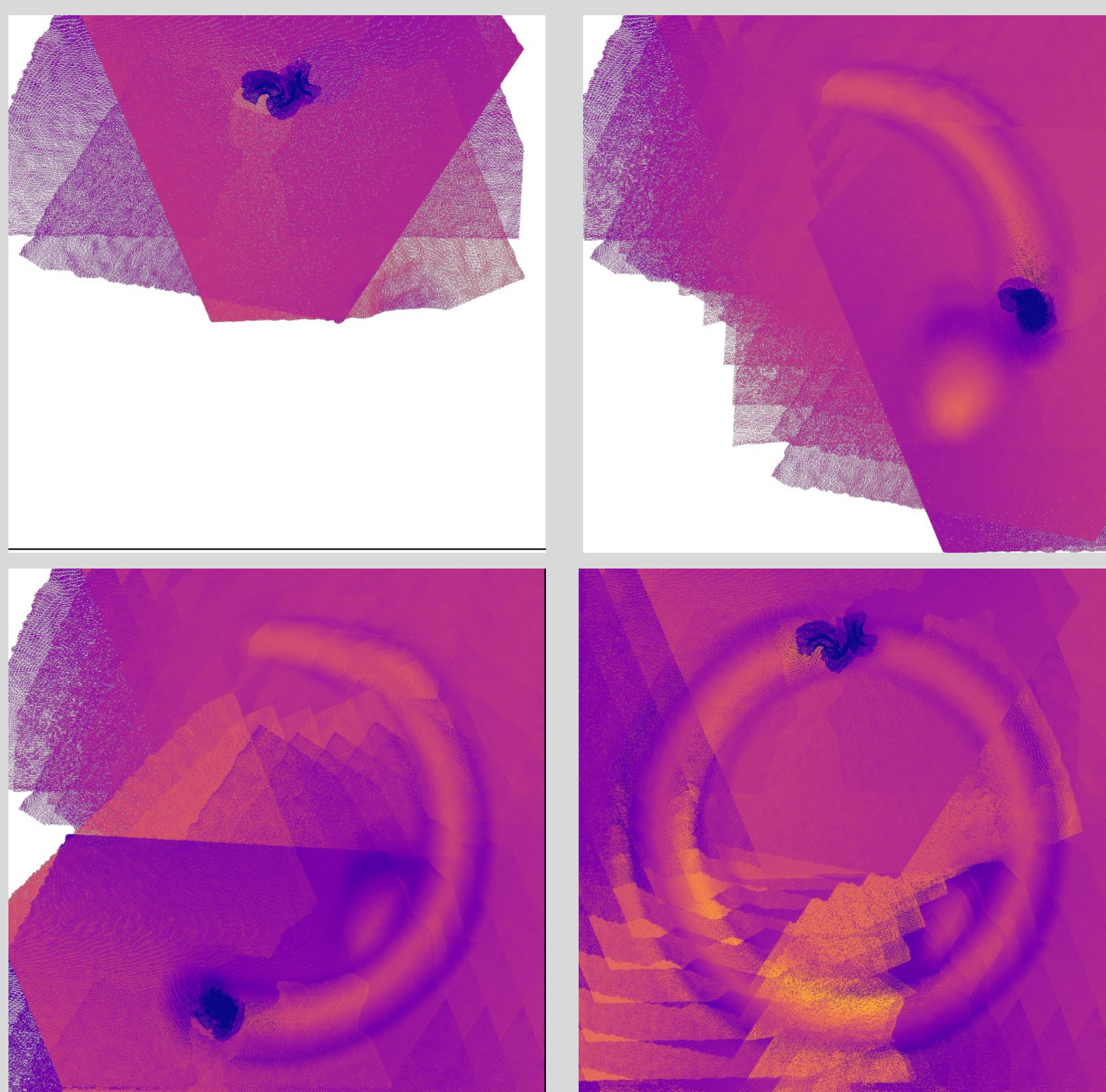
- Turns a point cloud into an easily computable matrix
- Each cell encodes height information
- Lowers computation and memory storage requirements



A feature transformed into a heightmap

Unfiltered multiple camera results

- Simultaneous displacement and mapping



Dynamic surface mapping

- White space represents unknown terrain

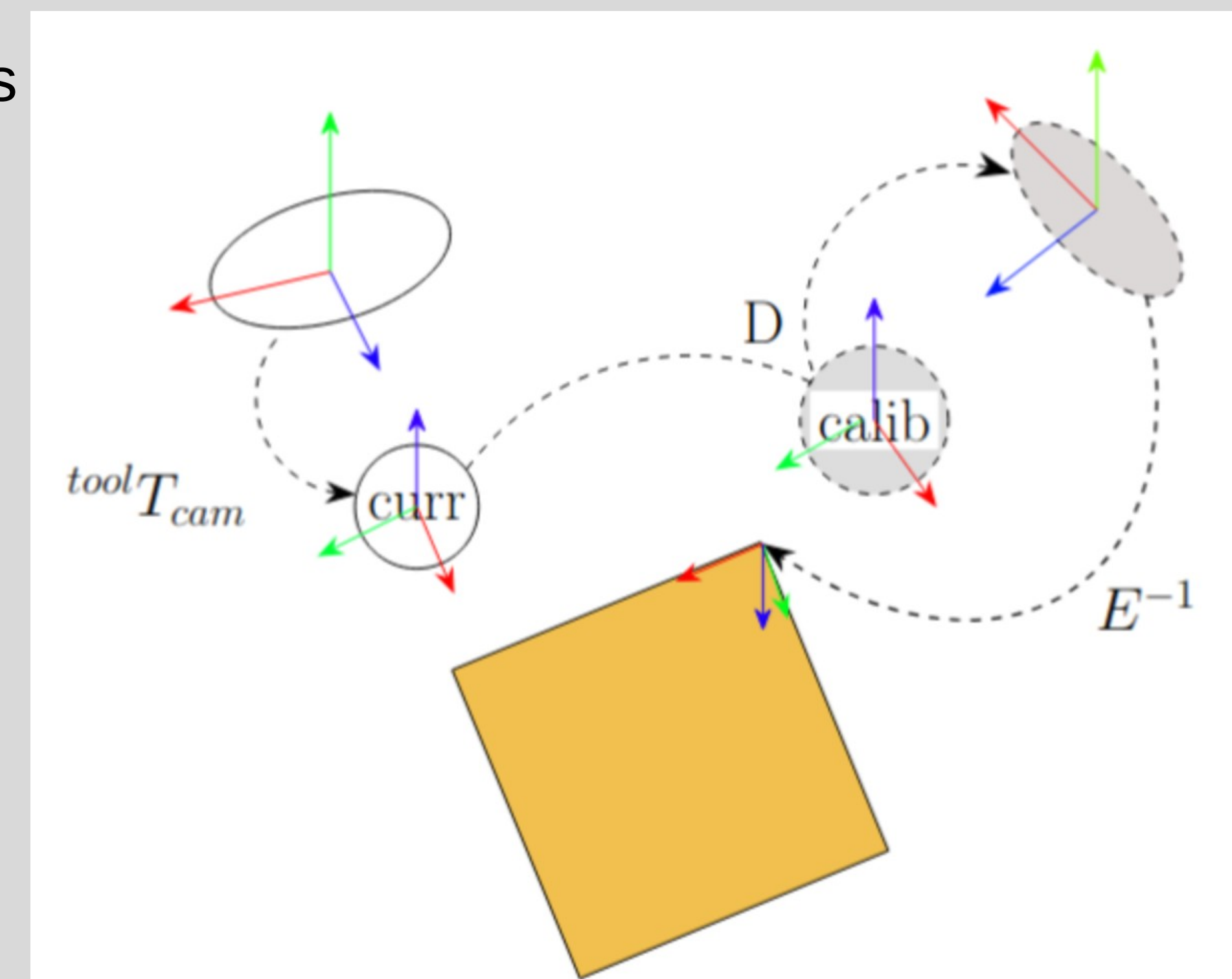
- As the tool traverses and deforms the terrain, the circular deformation can be clearly seen

Transformation

- RGBD capture point clouds (p_{sense}) that need to be transformed into the workspace (p_{task})
- By calibrating with a Charuco board, the camera extrinsics E and ${}^{\text{tool}}T_{\text{cam}}$ can be calculated

$$P_{\text{task}} = E^{-1} D {}^{\text{tool}}T_{\text{cam}} p_{\text{sense}}$$

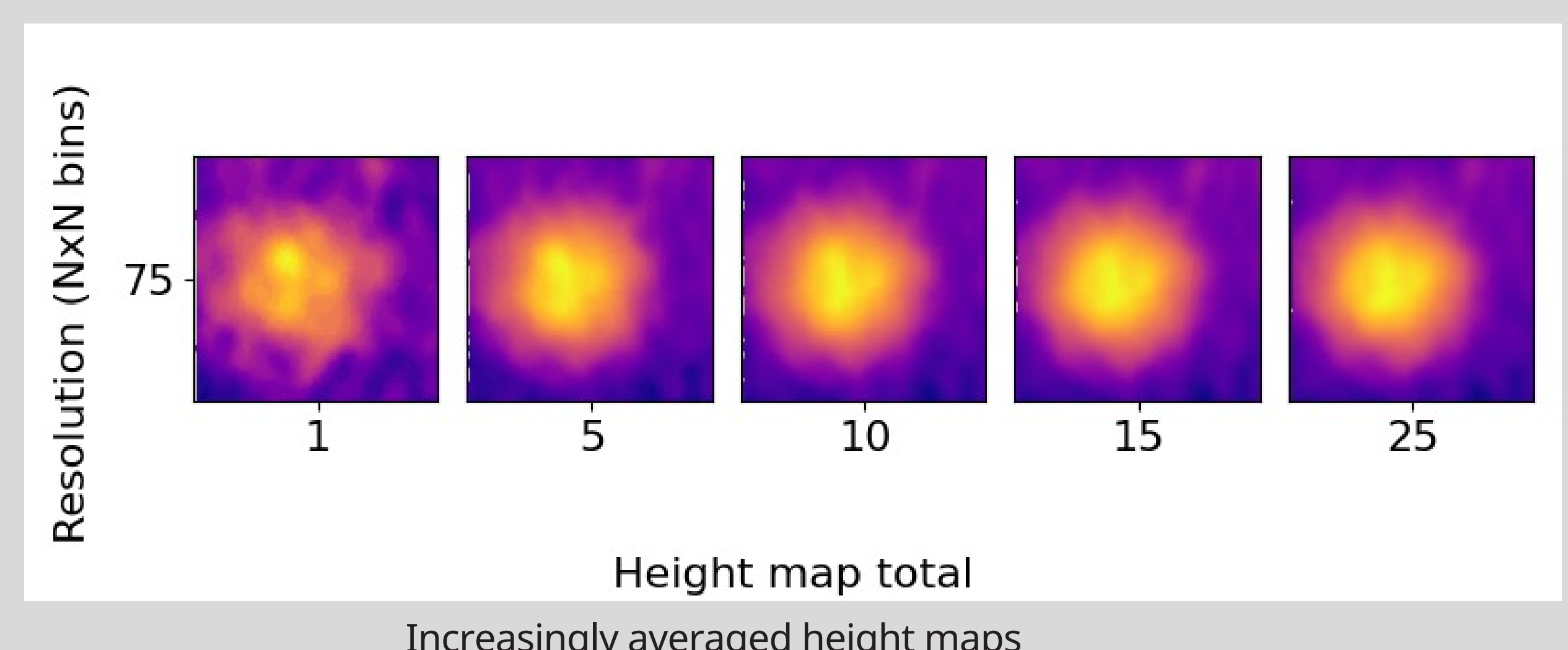
Where D is the transformation from the current camera position to the calibration position



Point cloud transformation

Solo camera

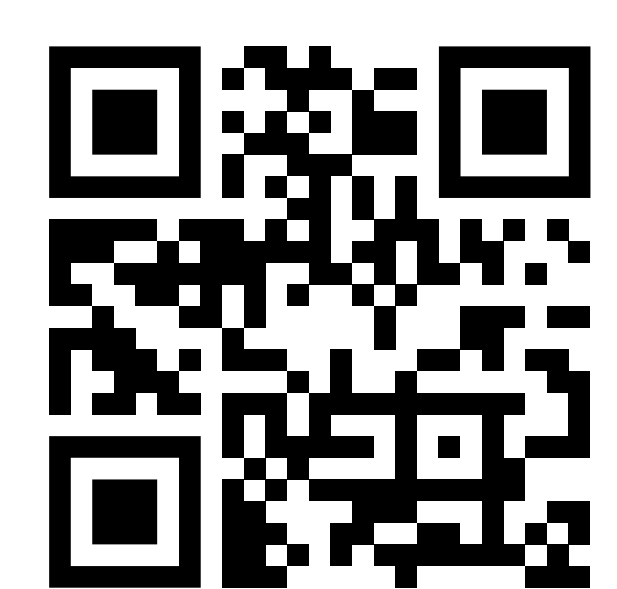
- Single camera directly above the feature
- Compared at various distances with industry standard LiDAR scanner
- At further distances, time-based height map averaging is required to improve accuracy



Height map total
Increasingly averaged height maps

Future work

- Fine tuning the alignment in real-time using ICP
- Collecting datasets which contain the force-displacement history of the terrain
- Methodology assessment for automated earthmoving



SCAN TO
DISCOVER MORE

1: $E, D, {}^{\text{tool}}T_{\text{cam}}$ are 4x4 homogenous transformation matrices, and points p_i are 4x1 column matrices